

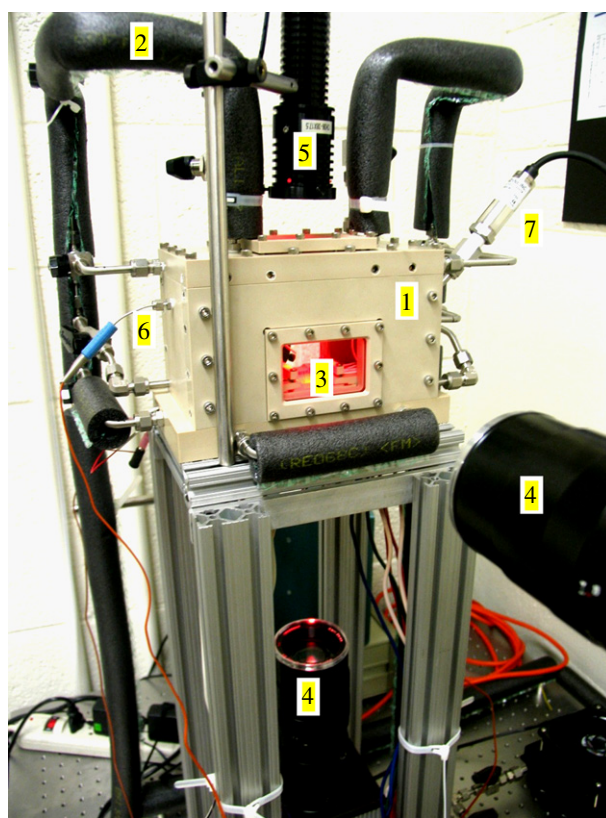


Fig. 4. Photo of experimental facility showing: (1) pool chamber, (2) secondary thermal loop, (3) test piece, (4) high speed cameras, (5) synchronized LED backlight, (6) pool thermocouple, and (7) pool pressure transducer.

3.4. Imaging and instrumentation

Two high-speed cameras were used to image the entire heater strip at 4000 frames per second (fps) with an exposure time of 1 μ s. For the side view, the cameras were set at a resolution of 1024×512 pixels, while for the top view, the resolution was 1024×256 pixels. Both cameras were equipped with macro-lenses (Carl Zeiss Makro 100 mm, 2 $\times$) set at a focal length of 100 mm. The spatial resolution was $34.3 \mu\text{m}/\text{pixel} \pm 0.1$.

Backlighting for the bottom and side views was provided by two red LED light sources (HPLS-36) pulsing alternately. The two high-speed cameras utilized a frame grabber (Berkeley Nucleonics model 6600) for data acquisition (proportional to image area). The pulse width was varied within $\pm 2 \mu\text{s}$ for best contrast to capture the surface and boiling characteristics.

The pool temperature was monitored by a thermocouple attached to the terminals near the test piece. A Agilent 34970A data logger recorded the temperature accurate to within 0.5°C . The pool pressure was measured by a digital pressure transducer (Omecon model 1000) with a resolution of $102,700 \pm 1400$ Pa. Heat transfer was provided by a Thermax 1000 circulating heater/chiller. The test fluid was a 50/50 mix of water and ethylene glycol maintained at 61 and 67°C in order to maintain a constant temperature and temperature, compensating for heat losses.

The electrical power to the test piece was provided by a JQE DC power supply capable of 1000 W and 1.5 A. All electrical measurements were recorded by a data acquisition system. Voltage and